Oslo, Norway
April 17 2026

Dear Reviewers,

We most sincerely thank the reviewers for their feedback and constructive comments. We have substantially revised the manuscript in response, and we believe that the feedback has led to a significantly improved manuscript. Our detailed responses follow below and in the subsequent pages. The reviewers' comments are included in italics, while our responses follow in roman text. Our responses are typically split into a **Comment** (responses to the reviewer) and an **Action** indicating the concrete changes made in the revised manuscript.

We also refer to the manuscript version with changes from the previous revision marked (additions only, deletions not marked). For many of the points below, our revisions are extensive and cover several sections and different parts of the manuscript. In these cases, we refer to the manuscript with changes marked, and specific sections therein, rather than including line numbers or reproducing the changes verbatim in this response letter.

In addition to our changes in response to the reviewer comments, we have revised the Abstract, Introduction, subheadings in Methods and Results, and references to the Supplementary Information to comply with the Editorial Policies.

We note that changes in the abstract and title are not included in the PDF with changes marked, nor are automated changes in the numbering of the references. We have replaced the word "molecular" by "solute" in the title, the revised title is:

"In-silico solute transport via perivascular networks in the human intracranial space"

Moreover, the revised abstract is as follows (additions/edits marked in blue):

"The mechanisms of intracranial solute transport are fundamental to human brain health. However, our understanding of their interplay remains incomplete, in part due to the complexity of integrating insights across scales, between species, and from different modalities. In this work, we combine mixed-dimensional mathematical modelling, multi-modal magnetic resonance images, and high performance computing to present a computational framework that predicts the spatiotemporal spreading of a solute across the subarachnoid space, ventricular system, and brain parenchyma, including surface perivascular spaces. To illustrate applicability, we simulate a series of scenarios relating to intrathecal tracer injections, including variations in cerebrospinal fluid production, intracranial pulsatility, perivascular fluid flow and -structure. This openly available framework allows for evaluating the integrated effect of different transport mechanisms, bridging from the vascular mesoscale to the organ-level macroscale and across time scales of seconds to hours, in subject-specific geometries."

During our work on the revisions, we also discovered one error in our perivascular flow model code. We have corrected this bug, rerun all simulations, pushed the updated software to our openly available GitHub repository, and will archive the final code version via Zenodo.

Reviewer #1

Reviewer #1 (Remarks to the Author):

I would like to thank the authors for their detailed responses to my comments and for the improvements they have made, in particular the enhancements to the usability of the computational platform for third parties.

Main Comments:
My main criticism of the original manuscript was that it attempts to straddle the line between a methods paper and a biomedical paper, and in doing so fails to meet the expectations of either. I had hoped that the revision would firmly reframe the work as a robust methods paper, highlighting the computational framework as the central contribution. Instead, the revision seeks to patch over the cracks caused by this straddling, without addressing the underlying issue.

In my view, the true value of the work lies in the computational framework, which is indeed a substantial and commendable achievement. Such a framework can be used to build a wide range of models, each designed to test or address specific hypotheses. By contrast, the simulations presented here should be seen only as illustrative demonstrations of the framework's potential, not as contributions to our understanding of neurobiology. The revised manuscript, however, continues to present them as biological findings (e.g., "In conclusion, our findings [...] reconcile seemingly conflicting observations in particular relating to directionality of perivascular flow [...]").

While the authors have added a comparison to the recent MRI work by Hirschler and colleagues, lending plausibility to their simplified approach, it remains a very limited and indirect form of validation and does not amount to a substantive test of the simulation results. Therefore, the gap between methodological contribution and biological interpretation is unresolved.

Comment: We thank the reviewer for their review, for their constructive comments and for reiterating and clarifying their concerns.

We believe that we now understand the reviewer's original and current comments, and are in a better position to address the underlying issues. In this revision, we have reframed our work in line with the comments from the reviewer. We emphasize the computational framework as the central contribution, clearly present model limitations, and have removed the claims pointed at by the reviewer as unsupported biological findings.

Action:
To address this main criticism of the reviewer, we have edited the entire manuscript substantially. Our changes are as follows.

- We have revised the abstract in line with the reviewer's comments, now highlighting the computational framework and its potential rather than specific biological interpretations. The changes in the abstract are shown on the first page of this letter.
- We have substantially revised the Introduction. We kept the first two motivating paragraphs, but revised the presentation of the key contributions and results (last paragraph of the Introduction) in order to reframe the work and results as suggested by the reviewer.
- We remark that we have changed the baseline model and as a consequence all model variations in response to comments from Reviewer #3. All reported numerical results and all subfigures have been recomputed and updated in the manuscript.
- We have revised all subheadings within Results
- We have edited all Results sections to address the reviewer's comments above, some sections extensively. A summary of the main changes is as follows.
    - In the revised manuscript, we now begin Results by a separate subsection, titled "A computational framework for intracranial solute transport", dedicated to describing the computational framework. This subsection is accompanied by a new Figure 1.
    - The baseline intrathecal tracer simulations are now presented as a first demonstration of this computational framework in the separate subsection, titled "In-silico tracer distribution after intrathecal injection". This section is supported by the revised Figure 2 (old Figure 1).
    - We have reframed the subsection formerly titled "Reduced CSF pulsatility strongly shifts intracranial enrichment patterns", now titled "Modelling CSF flow and dispersion induced by pulsatility". We have substantially revised this section and revised the associated Figure 3 (old Figure 2). In connection with this part, we also refer to the changes in Discussion as described below.
    - We have reframed and revised the subsection formerly titled "Perivascular flow shapes and accelerates molecular transport", now titled "Modelling net flow in human surface periarterial networks", in particular avoiding any claims of biological findings.
    - We have reframed and carefully revised the subsection formerly titled "Structural versus functional compartmentalization of perivascular spaces", now titled "The diffusive permeability of the PVS-CSF interfaces", in a similar manner as the preceding subsection. We have shortened the description of the model variation results in order to reduce emphasis in line with the reviewer's suggestion, and again avoid any claims of biological findings.
    - We have also reframed and made minor changes in the subsection formerly titled "Enlarged PVSs in the SAS delay periarterial and intracranial tracer enrichment", now titled "The width of the PVS".
- We have carefully revised the Discussion, removing biological interpretations as appropriate throughout, extending and rewriting the limitations part, and revising the conclusions. Just to give a few examples, we have removed phrases such as: "All in all, we interpret these results as strongly indicative of that PVS flow driven by CSF production and cardiac peristalsis alone is insufficient to explain the rapid perivascular transport observed in humans", "In conclusion, our findings [...] reconcile seemingly conflicting observations in particular relating to directionality of

perivascular flow [...]", and "[...] thus contributing to quantifying the properties of the human PVS-CSF interface".

- Overall, we have carefully revised any claims made relating to biological findings, clearly framing our observations as model predictions. This includes, but is not limited to, the specific text examples pointed out by the reviewer in this and in the previous review.

Considering the extent of our edits, we refer to the manuscript with changes marked instead of including the line numbers of the changes verbatim here.

Further Comments:
Terminology ("enrichment"): While "molecular enrichment" has been removed, the manuscript continues to use the term "enrichment." This is not common or accepted terminology in the field, even though it is apparently used in studies of Eide and colleagues. It would be clearer to use established physical terminology, given that the manuscript will primarily reach a physics-affine readership.

Comment: We now understand that the reviewer was critical not only to the use of the term "molecular enrichment" but to the term "enrichment" itself. **Action:** In response, we have iterated through the manuscript and the supplementary information and removed the use of the term "enrichment". We have replaced this term as appropriate e.g. using "tracer distribution" or "appearance of tracer".

Glymphatics: The authors set advective interstitial flow to zero, which by definition excludes glymphatic transport. Nevertheless, the manuscript continues to make extensive reference to glymphatics. This seems more aimed at appealing to a biological readership than an accurate reflection of what the simulations capture. The likely readership of this manuscript is physics-affine rather than biological, and the terminology should reflect that.

Comment: We had used the term "glymphatic MRI" to refer to the clinical protocol used by Eide and colleagues (as this is the term introduced and used by them). We understand that this terminology can be inaccurate, as pointed out by the reviewer. **Action:** We have therefore replaced the term "glymphatic MRI" by the term "contrast-enhanced MRI" in the revised manuscript and supplementary information.

Overinterpretation of ventricular and cisterna magna results: The ventricular and cisterna magna simulations remain unvalidated against higher-dimensional simulation data. Although the authors acknowledge these limitations, they nevertheless draw conclusions that extend beyond what the model can robustly support. This overinterpretation is especially problematic when results are linked to intrathecal drug delivery or human pathophysiology.

Comment: As described in our response to the main criticism by the reviewer, we have carefully revised the entire manuscript, removing claims of biological findings, clearly framing our results as model predictions, and clarifying the limitations of the framework.

The reviewer refers to “higher-dimensional simulation data” for the “ventricular and cisterna magna simulations”. We assume that the reviewer refers to the flow simulations in the CSF spaces (including the ventricular system and cisterns) here. Our simulations of flow in the ventricular system and cisterna magna at systole and peak respiration capture the three-dimensional geometry of these spaces. We do not capture the time-dependent response or other dimensions associated with e.g. uncertainty. If these are the higher dimensions referred to by the reviewer, then we absolutely agree and acknowledge that our flow-simulation-based dispersion estimates are not compared against spatio-temporal simulations.

Action:

- The reviewer’s comment regarding drawing “conclusions that extend beyond what the model can robustly support” has been addressed by the revisions in response to the main criticism by the reviewer (as described above).
- In particular, the subsection now titled “Modelling CSF flow and dispersion induced by pulsatility” has been reframed and substantially revised.
- In response to this comment from Reviewer #1, and the comments from Reviewer #3, we have substantially revised the description of limitations in the Discussion. This includes an extended discussion of this point.

Conclusion: This could be a wonderful methods paper if the authors chose to remove claims of biologically relevant findings. As long as the manuscript maintains its straddled form, I do not find it suitable for publication.

Comment: We note and appreciate the reviewer’s comments and conclusion. As described above and as shown in the manuscript with changes marked, we have revised the manuscript substantially in order to address the reviewer’s comments – including choosing to remove claims of biologically relevant findings. We sincerely hope that we have interpreted the reviewer correctly and that our changes address their concerns.

Reviewer #1 (Remarks on code availability):

No further comments on the code.

Reviewer #2

Reviewer #2 (Remarks to the Author):

I'm completely satisfied with the authors' responses.

Reviewer #2 (Remarks on code availability):

I looked at the github and it seems fine but I did not run the code.

Comment: We thank the reviewer for examining the revision and the positive feedback.

Reviewer #3

*Review on the revised version (Revision 1) of "**In-silico molecular transport via perivascular networks in the human intracranial space"** by Marius Causemann, Miroslav Kuchta, Rami Masri, and Marie E. Rognes.*

First, I would like to thank the authors for their answers to my comments, and the associated clarifications. Some were very useful, but some reinforce some of the concerns I had in my initial review (as detailed below). I would also like to underline that, in my opinion, the changes made in the manuscript do not always fully reflect these answers. Thus, while clearly improved (see, e.g., restructuration of the methods and SI with clearer separation between flow and transport), this revised version is still not sufficiently clear on the limitations of the modeling choices and their impact on the interpretation of the simulated physio-pathological behaviors. The general context and wording are still very broad and suggest relevance not only for intrathecal injection of tracers but also for brain clearance in the context of neurodegenerative diseases (e.g., the term clearance is still to be found more than 30 times in the MS, despite my initial comment #1). As already noted in my initial review, there is an unusual amount of misunderstanding and controversy in the field of brain clearance. Thus, I still think the authors should adopt a clearer and more careful presentation.

Comment/Action:
We again thank the reviewer for their efforts and highly constructive feedback on our manuscript. We give an overview of our response and revisions here first, and go into more detail point-by-point below.

The reviewer emphasizes that the current manuscript is not sufficiently clear on (1) the limitations of the modelling choices, and (2) their impact on the interpretation of the simulated behaviour. In addition, the reviewer asks for (3) a clearer and more careful presentation. We have revised the manuscript substantially in line with these comments. We have also revised the manuscript in response to the comments from reviewer #1, with which there is significant overlap.

In short, we have addressed (1)--(3) by the following major revisions.

- We have completely revised the description of limitations (more details below)
- We have completely reframed the main contributions and results of our work. Here, we refer to our responses to the main criticism from reviewer #1 for an overview of the changes made and the manuscript with changes marked for the precise changes.
- We have carefully changed the wording of the manuscript in response to the reviewers' comments regarding terminology.

Importantly, we have also revised the computational models, rerun all simulations, and updated all simulation results.

Regarding the term clearance: In our modelling framework, the solute concentration is

allowed to leave the intracranial compartments through the outer boundaries. We have referred to this as solute clearance in the original manuscript, but note that the reviewer is critical to the use of the word clearance. We therefore refer to this as solute efflux in the revised manuscript. In the revised manuscript, we refer to clearance three times: once in the Introduction, once in the context of perivascular pathways, and once in the Discussion in connection with previous work explicitly referring to brain clearance in connection with lifestyle factors.

We emphasize that it has at no point been our intent to not reflect our answers by changes in the manuscript. Please do let us know if there are any further instances where this is the case, and we will revise accordingly.

Major concerns

A. Interface conditions between PVS leaf nodes and parenchyma (related to my initial comment #12a).

In response to my initial comment #12a, the authors now clearly state they consider the velocity to be null in the parenchyma (uPAR(x,t)=0 in \OmegaPAR) in all equations relative to tracer transport (Line 400). In their response to this same comment, they also clarify that the harmonic extension of the SAS pressure field in the parenchymal domain is not used to compute a parenchymal velocity akin to a Darcy velocity but to extrapolate the pressure field over the segments of the PVS network embedded in \OmegaPAR.

Comment/Action: The reviewer's previous and current concerns in connection with the CSF flow modelling prompts us to make one important clarification. Our key modelling contribution in this work is to present an integrated model of solute transport in the three-dimensional intracranial compartments and embedded, topologically one-dimensional perivascular spaces. It has not been our intention to present a complete, integrative model of CSF and ISF fluid flow in these compartments. We now explicitly state this in the revised manuscript in the Results section titled Modelling net flow in human surface periarterial networks:

"We remark that while we present a coupled framework for solute transport in the CSF spaces, brain and surface PVS networks, a comprehensive model of CSF and ISF flow is not within the scope of this work. In particular, we here demonstrate using separate; i.e., not coupled, flow models in the CSF and PVS."

Our rationale for this is, in part, the lack of consensus and significant debate surrounding the precise anatomical and functional nature of fluid exchange between the PVS, the SAS and interstitium, e.g. within surface and penetrating perivascular spaces, and between the PVS and surrounding CSF spaces and/or neuropil. As the reviewer also points out below, there are several hypotheses, but each come with their own implications, and we consider complete models of each of these to be beyond the scope of the current study. The significant mathematical, numerical, and computational complexity of a fully coupled CSF/ISF/PVS flow model contributes to this consideration, as do the incompleteness of our data on the vascular networks.

What we are presenting are demonstrations of how one can apply the computational framework for solute transport to investigate the isolated effects of variations in the CSF/ISF flow in separate compartments, e.g. with higher surface PVS flow.

*This pointed my attention toward an important ingredient in the modelling, which I think leads to inconsistent interface conditions at the PVS leaf nodes, as detailed below (**A.b** and **A.c**). To this end, I first comment on how this extrapolated pressure field is used as a boundary condition for computing the steady state PVS flow induced by CSF production in the ventricles (**A.a**), an important assumption that should be further justified by the authors.*

A.a. Steady fluid flow in the PVS network induced by CSF production: *L361-368 now clearly state that the boundary conditions for this flow are pressures imposed at the end nodes of the PVS network. These pressures are chosen as to be locally equal to the pressures in the adjacent CSF space in the SAS (or its harmonic extension in the parenchyma). However, even if unresolved in the data used by the authors, these PVS extend further and, in the present work, they are considered to be fully impermeable to fluid flow (see my initial comment #4). Thus, I don't see how there could be continuity of pressure between their end nodes (which are once again fictitious end nodes due to the lack of resolution in the imaging) and the pressure in the adjacent SAS. The authors need to justify this modelling choice.*

Comment: We agree with the reviewer that this is an important point and that there were inconsistencies in our modelling choices here. We fully agree with the reviewer that our PVS network end nodes are fictitious ends. Upon further consideration, we also agree that imposing continuity between the PVS end node pressures and the CSF/parenchymal pressure represent an inconsistent modelling choice. Specifically, such a pressure-induced flow model would assume direct fluid contact between the PVS and the surrounding CSF/tissue at the network ends, which is in contradiction with the assumption that the network ends are fictitious end nodes.

Thus, rather than justifying this modelling choice, we have decided to remove the steady fluid flow in the PVS network induced by CSF production from the baseline model and all simulation scenarios. We now treat the CSF flow in the SAS and the PVS flow as fully (instead of partially and inconsistently) decoupled. As the pressure-driven PVS fluid flow was one to two orders of magnitude smaller than the other contributions to the PVS flow, we note that the difference in the reported results is minimal.

Just to be very clear: we do not consider the PVS "to be fully impermeable to fluid flow" in case that is what the reviewer indicates here. We do allow for fluid flow within the PVS, and for fluid flow in and out of the PVS network ends. However, we do not consider a coupled model of fluid flow between the CSF spaces and the parenchyma, nor between the PVS network ends and their surroundings. We fully agree with the reviewer that this is a clear limitation, and we emphasize this point again in the revised limitations section of the Discussion.

Action:

- We have removed the steady fluid flow in the PVS network induced by CSF production from the baseline model, reran all simulations, updated Methods, all Results and Figures accordingly. We refer to the manuscript with changes marked for the detailed changes.
- We have substantially revised the description of limitations in the Discussion, as requested by the reviewer. We detail these revisions in our response to the reviewer's point **D** below.

A.b. Interface condition for the flow problem: *As a result, there is an inconsistent interface condition for the flow problem (mass conservation of CSF/ISF) at PVS parenchymal leaf nodes. On one hand, the velocity at the PVS end nodes is non-zero (as acknowledged by the authors: Lines 386-388: We remark that […] fluid entry and exit is allowed at the end nodes of the PVS networks). On the other hand, the velocity in the parenchyma is not solved for and considered to be zero. As a result, mass conservation for CSF/ISF cannot be fulfilled at these points,* ***which is not acceptable from a fundamental point of view and has a strong impact on the transport problem*** *(****see A.c****). In my opinion, there necessarily needs to be an additional ingredient to ensure mass conservation of CSF/ISF at parenchymal leaf nodes, e.g., retrograde CSF flow between other anatomical layers along the PVS (Intramural Peri-arterial Drainage Pathway hypothesis) or bulk flow within the parenchyma (glymphatic hypothesis).*

Comment: The reviewer is absolutely correct in that we consider models of CSF flow in the surface PVS that have non-zero net flow at the end nodes, and that we do not account for any coupling between this flow and fluid flow in the 3D domains (neither in the SAS or the parenchyma). Indeed, we do not consider coupled models of fluid flow between the CSF spaces and the parenchyma, nor between the PVS network ends and their surroundings. (We do remark that our estimates of net flow in the PVS network itself are mass-conservative in the sense that the fluid volume that flows into the PVS networks also flows through and out of the networks.)

As described above (just before **A.a**), our rationale for this is three-fold: the lack of consensus regarding the underlying physiology, our lack of imaging data for the deeper vascular networks, and last but not least, the mathematical, numerical and computational complexity associated with introducing a fully coupled CSF/ISF/PVS flow model. Indeed, such a model would entail a number of additional modelling choices. After truly careful consideration (including prototype implementations), we have concluded that a fully coupled CSF/ISF/PVS flow model is beyond the scope of the current study. Instead, we argue that our revised decoupled approach is the most transparent yet feasible solution.

Note that we absolutely agree with the reviewer's physical and biological premise: in reality, fluid is likely to be exchanged between the PVSs, parenchyma and SAS. However, because our network end nodes are artificial truncations, enforcing a local mass balance at these specific coordinates would require us to introduce localized, artificial sinks/sources in the parenchyma and SAS. Doing so would effectively hard-code a specific (and highly debated) fluid pathway into the model at arbitrary geometric locations.

Action: In light of our comments above, we now explicitly state that we do not consider a

coupled model of fluid flow in the CSF spaces/parenchyma and the PVS networks in Results (as marked in blue above), in Methods (in the section titled Perivascular fluid flow induced by arterial wall motion) and in the Discussion. We refer to the manuscript with changes tracked for the precise changes. Moreover, we have substantially revised the limitations section of the Discussion, as detailed in our response to the reviewer's point **D** below.

A.c. Interface condition for the transport problem: *Despite the non-zero velocity at the PVS parenchymal leaf nodes, the authors impose a no-flux boundary condition for the transport problem at these same end nodes (L408-409). Thus, tracer is transported to the end nodes by advection (relatively fast) but "the network end nodes represent barriers for molecular efflux into or out of the perivascular network and […] all solute exchange between the PVSs and their surroundings takes place via the lateral outer PVS surface and is regulated by the exchange parameter (SI L182-183)". This results in* ***artificial tracer buildup*** *(L423-424) there (Footnote 1: L422-425 "To account for the unresolved continuation of the perivascular network beyond the imaging resolution limit, we implemented a permeability gradient to avoid artificial tracer buildup. At the terminal leaf nodes of Λa, permeability was increased by a factor of 100 to simulate tracer exchange with the surrounding CSF/tissue, and linearly decreasing to the baseline value at the adjacent network node"). Increasing the permeability of the PVS lateral surfaces* ***by a factor 100*** *(L424) is then a trick that enables* ***tracer exchange with the surrounding CSF/tissue*** *(L425). This in turn artificially builds up a tracer concentration gradient between the root and leaf nodes of the arterial PVS network, which is necessary for diffusion and dispersion to be effective throughout this network.*

A similar argument can be made for the dispersive contributions as the enhancement factor scales with the velocity squared.

My opinion on this is that these modelling choices are too rough, and in practice undermine the physical validity of the results. I am not convinced by the author's interpretation of this issue as a problem of resolution, as suggested by the following text in the initial MS version: "L373-376: At the terminal leaf nodes of Λ, permeability was increased by a factor of 100 to simulate tracer exchange with the surrounding CSF/tissue, and linearly decreasing to the baseline value at the adjacent network node. The same contradictions as described above would remain even for a PVS network acquired at sufficient resolution to identify the PVS spaces along penetrating arterioles and ascending venules in the parenchyma.

Comment:
We thank the reviewer for this astute and constructive comment. We agree that this approach (to impose a no-flux end node condition in combination with a 100x diffusive permeability multiplier along the PVS end segment) has a number of limitations. Our target was to facilitate exchange between the PVS network and the surroundings near the network ends in a reasonably simple manner.

To address this issue, we have completely removed the artificial permeability multiplier, and substantially revised our mathematical model. We have now extended our framework to a mixed 3D-1D-0D coupling, where we model the unresolved perivascular continuations as individual 0D compartments attached to each arterial leaf node. This approach can be

viewed as a reduced version of the approach introduced by Fritz et al (2022)[1]. These 0D compartments are assigned effective volumes and surface areas derived from a branching tree assumption (Murray's Law), representing the PVS down to the precapillary level. The 0D compartments exchange mass with the 1D network via Robin-type interface conditions equipped with proper upwinding, and subsequently exchange mass with the surrounding 3D domain (parenchyma and CSF) via a localized volumetric source term.

To transparently demonstrate the impact of this change, we have added a comprehensive comparison in the Supplementary Information. In the revised Supplementary Information, we explicitly compare the old multiplier approach (testing values from 1x to 1000x) with the new 0D model under the high PVS flow / vasomotion regime. As detailed in the supplementary text, we demonstrate that the macroscale distribution of tracer and upstream metrics such as first-time tracer arrival are robust to the treatment of the network leaf nodes. However, the old model features unphysiological concentration spikes at the arterial leaf nodes, even with high multiplier values. These local artifacts are successfully resolved by the new 0D approach.

Actions:

- We have removed the 100x permeability multiplier from the model entirely.
- We have updated the governing equations in the main text and Methods to include the new 3D-1D-0D coupled transport model.
- We have completely rerun all simulations using the new 0D compartments and updated all figures in the manuscript.
- We have added a detailed derivation of the 0D compartments to the Methods section.
- We have added a comprehensive sensitivity analysis and comparison between the old multiplier approach and the new 0D model to the Supplementary Information to transparently present the updated model (Supplementary Figures S7 and S8).

A.d. *What is the justification for the choice of a factor 100? What is the impact on the tracer dynamics?*

Comment/Action: We thank the reviewer for raising this point. To be honest, the factor of 100x was originally chosen as a heuristic compromise. It was large enough to reduce the non-physical tracer accumulation at the PVS networks leaf nodes, but lacked a physical justification.

To explicitly answer the reviewer's question regarding its impact on tracer dynamics, we have conducted a comprehensive sensitivity analysis, which is now included in the revised Supplementary Information (see also our answer to the previous comment). Because the ad-hoc multiplier could not fully resolve the local artifacts and lacks physical justification, we have completely removed it from the model. As detailed in our response to the previous comment, we have replaced it with a new 3D-1D-0D coupled model, which naturally handles

[1] Fritz, Marvin, et al. "A 1D–0D–3D coupled model for simulating blood flow and transport processes in breast tissue." *International Journal for Numerical Methods in Biomedical Engineering* 38.7 (2022): e3612.

downstream drainage into the surrounding tissue, and is parameterized using estimated values based on branching tree assumption (Murray's Law).

Could the signal measured in PVS around larger veins be compared with clinical measures for validation? In fact, this veinous dynamics integrates all contributions of the different ingredients and is almost never commented in the manuscript, while it is commented for arterial PVS in great detail (Fig. 4E and associated comments).

Comment:
The reviewer is correct that our manuscript comments extensively on the arterial PVS while largely leaving the venous PVS unaddressed, despite including it in our model. Unfortunately, comparing in-silico venous PVS signals with clinical measures is currently difficult. As recently noted by Eide and Ringstad (2024), veins at the human brain surface are generally much harder to depict on MRI than arteries, making perivenous enhancement difficult to capture in-vivo. They reported identifying perivenous tracer enrichment only in very few instances. From a modeling perspective, we included the complete perivenous network in our computational geometry for anatomical completeness, but we intentionally kept its physics simplified. Unlike the arterial network, we do not model any active fluid flow mechanisms (such as peristalsis from cardiac pulse waves or low-frequency vasomotion) in the veins. Transport within our venous PVS is currently driven by diffusion and passive exchange with the surrounding tissue alone. This lack of active venous flow modeling – combined with our simplified model of the parenchyma – is the primary reason we restrict our simulation timeframe to the first 24 hours. While our current framework is well-equipped to study the early influx and distribution phase (where CSF pulsatility and arterial PVS flows dominate), it lacks the detailed parenchymal and venous clearance mechanisms that would be required to accurately simulate later stage brain clearance (e.g., 48 to 72 hours post-injection).

Actions:
- As part of the rewrite of the limitations section of the Discussion, we have clarified the limitations of our simplified modelling or perivenous dynamics, and outlined our rationale for focusing on the early phases of the contrast-enhanced MRI procedure.
- In Methods, we clarify that perivenous transport is solely driven by diffusion in our model.

B. Structural versus functional compartmentalization of perivascular spaces (related to my initial comment #4).
The authors clarified in the MS that the change in compartimentalization is only considered in the transport problem, while the flow problem is always solved for fully compartimentalized PVS (L281- 282). Their rationale for this choice, as described in their answer, is as follows: 1/ they want to isolate the effect of the PVS outer interface permeability on molecular transport, 2/ there is a current lack of computational methods to quantify net PVS flow due to vascular pulsations in the presence of a permeable outer PVS interface over relevant time scales.

1/ I agree that from a theoretical point of view, a tube can be impermeable to fluid flow and

permeable to solute transport. However, the whole text about permeability is quite ambiguous, from definition (ξ is a permeability allowing for transfer/exchange across the interfaces between the PVS and their surroundings (L397-398)) to parametrization (see e.g. Legend of Table 1: We use the astrocytic endfeet permeability ξEF estimated by Koch et al. [92]). To be noted, in their paper, Koch et al. clearly distinguish between permeability for advective and diffusive transport. Here, the authors should at least use the term "diffusive permeability".

Comment/Action: We have revised the manuscript in accordance with the reviewer's suggestion, now using "diffusive permeability" in place of "permeability" when referring to this parameter.

They should take special care to clarify the relatively limited implications of this parametric study with regard to larger question of compartementalization, for example by removing the sentences that seem to imply they also consider situations in which penetrating periarteriolar spaces are continuous with the CSF space, e.g: "To what extent are surface PVSs in communication with the surrounding SAS and to what extent are these structurally separated compartments?" (L272) "Bakker and coauthors report that the subarachnoid space, the cisterns, ventricles and penetrating periarteriolar spaces form a continuous CSF space" (L274).

Comment/Action: We note and appreciate the reviewer's constructive critique. In order to take care and to clarify, we have (i) substantially revised the associated subsection of the Results (now titled The diffusive permeability of the PVS-CSF interfaces) and (ii) revised this entire paragraph of the Discussion, including but not limited to editing the two sentences pointed at by the reviewer. We consider it important to reference the findings from Bakker et al but we have revised the sentence, which now reads as:

On the other hand, Bakker and coauthors report that the SAS, the cisterns, and spaces surrounding major arteries and veins form a continuous CSF space [...]

They should also clarify conclusive sentences such as "Thus, fast transport along the PVS is not contigent upon a structural compartmentalization of the PVS, whereas a sharp concentration gradient between PVS and CSF is unlikely without a restricting barrier." (L187-189).

Comment/Action: In response to comments from both Reviewer #1 and #3's, we have removed claims of biological findings. In particular, in the revised manuscript we have removed the sentence pointed at by the reviewer here.

In fact, as soon as the walls becomes permeable for fluid transport (advective permeability), the transport problem becomes fully dependent of the flow problem and isolating the effect of the PVS outer interface permeability on molecular transport is meaningless.

2/ There are some studies in other contexts which look at the coupling between flow, transport and dispersion in tubes with permeable walls. Of note, even if only diffusive permeability is considered, there is already an impact on the dispersion coefficient, as the shape of the concentration profile is impacted at the microscopic scale.

Comment: We thank the reviewer for the pointer towards relevant studies, which prompted us to re-examine relevant literature. In addition to the work related to PVS dispersion referenced in the previous version of the manuscript, we note e.g.

Decuzzi, P., Causa, F., Ferrari, M., & Netti, P. A. (2006). The effective dispersion of nanovectors within the tumor microvasculature. *Annals of Biomedical Engineering*, *34*(4), 633–641. https://doi.org/10.1007/s10439-005-9072-6

Gentile, F., Ferrari, M., & Decuzzi, P. (2008). The transport of nanoparticles in blood vessels: the effect of vessel permeability and blood rheology. *Annals of Biomedical Engineering*, *36*(2), 254–261. https://doi.org/10.1007/s10439-007-9423-6

In both studies, the authors analyze the effective diffusion of solutes in capillaries with fluid permeable walls (advective permeability), and conclude that "In permeable conduits, the effective diffusion reduces significantly compared to normal non-leaky vessels [...]".

On the other hand, transport in tubes with walls that are permeable to the solute but not the fluid (diffusive permeability) was considered by e.g.

Zimmerman, R. A., Severino, G., & Tartakovsky, D. M. (2018). Hydrodynamic dispersion in a tube with diffusive losses through its walls. *Journal of Fluid Mechanics*, *837*, 546–561. https://doi.org/10.1017/jfm.2017.870

Dejam, M., & Hassanzadeh, H. (2021). The role of a porous wall on the solute dispersion in a concentric annulus. *Physics of Fluids (Woodbury, N.Y.: 1994)*, *33*(11), 116602. https://doi.org/10.1063/5.0070653

In the latter, the authors consider the effective diffusion/dispersion of a solute moving by advection and diffusion in a (long) concentric annulus surrounded by a porous medium/wall. The authors again report that "solute dispersion is reduced in the presence of the porous wall".

We also note the study

Romanò, F., Suresh, V., Galie, P. A., & Grotberg, J. B. (2020). Peristaltic flow in the glymphatic system. *Scientific Reports*, *10*(1), 21065. https://doi.org/10.1038/s41598-020-77787-4

While this study is placed in the context of flow in penetrating perivascular spaces surrounded by an elastic, fluid-permeable glial boundary, their analysis is also relevant for other fluid-permeable boundaries from a mathematical point of view. That being said, the authors do not provide estimates of the time-averaged net flow in the PVSs – which we would require for our modelling framework.

Action:

- We have added a sentence in the Discussion of the revised manuscript to underline

the potential effect of permeable walls on dispersion in the PVS, including two of the references listed above, reading as:

Moreover, the effective diffusion is typically reduced in conduits with semi-permeable compared to non-permeable walls, see e.g. the analyses [...] and references therein.

- We have revised the statement regarding "lack of computational methods" and have added a reference to the study of Romano et al (2020) listed above.
- As part of the rewrite of the limitations section of the Discussion, we have revised our description of the limitations associated with not accounting for a fluid-permeable outer PVS boundary, see more details in our response to point **D.** below.

C. Enrichment (related to my initial comment #2). *Despite changing the title, the authors chose to keep this terminology quite extensively in the manuscript, but without clarifying its definition. An explicit definition should be provided.*

Comment: We now understand that the reviewer was critical not only to the use of the term "molecular enrichment" but to the term "enrichment" itself. **Action:** In response, we have iterated through the manuscript and the supplementary information and removed the use of the term "enrichment". We have replaced this term as appropriate e.g. using "tracer distribution" or "appearance of tracer".

D. *As a result of the above comments, I think the limitation section of the discussion should be thoroughly rewritten. Besides the constraints associated to the limited spatial resolution for the PVS networks, which is emphasized on L304-305, the limitations about the underlying physics (e.g. additivity of enhancement factors, see my initial comment #14 that did not lead to any changes in the MS text) and boundary/interface conditions (see comments above) should be explicated much more clearly in a single section.*

Comment/Action: We appreciate the highly constructive suggestion from the reviewer. In response, we have completely rewritten the limitations section of the Discussion. Concretely, in this revised section (spanning five paragraphs) we now discuss, in order:

- Limitations associated with the flow modelling (lack of coupling between the flow in the CSF spaces/PVS, PVS assumptions)
- Limitations associated with the dispersion modelling in the CSF spaces, including the simplified flow models and additivity of enhancement factors.
- Limitations associated with the flow modelling within the parenchyma, and our handling of the solute exchange at the PVS network ends.
- Limitations associated with limited data resolution and other limitations

Moreover, specifically regarding the reviewer's initial comment #14, we realize that we failed to make the intended changes in the manuscript in the previous revision. Our apologies. In the revised limitations section, we now state that:

In addition, we combine cardiac and respiratory dispersion enhancement factors in an additive manner. While there are theoretical considerations indicating that contributions

from different frequencies are indeed additive [...], we note that such derivations rely on several simplifying assumptions, and therefore consider our dispersion estimates as heuristic.

Specific comments

- Fig. 1C. *Please specify total mass of injected tracer and injection duration on the figure.*

Comment/Action: We have updated the figure as suggested, now specifying the total mass of injected tracer and duration on the figure.

- L92: *We assume a constant CSF inflow. Please link this assumption to CSF production.*

Comment/Action: To link the assumption the CSF production, we have updated the sentence as follows: "To model the convective flow field, we impose a constant CSF inflow of 400 mL/day at the lateral ventricle surfaces, representing the average daily physiological CSF production rate".

- L200-201: *In relationship to my comment Aa above, I am concerned that these highly unrealistic boundary conditions translate into inconsistencies into the patho-physiological behaviors of the system induced by PVS enlargement in the SAS. Here, because of the reduced effective resistance of the PVS network, but the hypothesis of a constant pressure gradient, the steady flow velocity increases (L200-201). The alternative asymptotic limit of constant flow and reduced pressure gradient could also have been considered.*

Comment/Action: We understand the reviewer's concern. We could indeed have considered a reduced pressure gradient as suggested by the reviewer; however, as we no longer include the PVS flow induced by a constant pressure gradient (see our response to the reviewer's comment **A.a)**, we consider this alternative to be less relevant. Moreover, we have reframed the description of this simulation scenario, now emphasizing "illustrate how these types of questions could be addressed within our modelling framework". We hope that these two points address the reviewer's concern.

L234-236: For validation purposes, the authors compare the proportion of ***tracer amount*** *entering the parenchyma in the simulations with the ratio of* ***peak concentration values*** *in parenchyma versus SAS reported in a MR human study of intrathecal gadobutrol injection (L234-236[2]) (Footnote 2:* L234-236 We find that about **20% of the tracer reaching the cranium enters the parenchyma**, which aligns with the **reported peak concentration values of 0.5mmol/l in the SAS and 0.1mmol/l in the parenchyma** reported by Watts et al [68].*), without accounting for the different volumes of the respective compartments nor for the different time scales (average over whole simulation duration if I understand correctly versus peak values). This adds up to the confusion.*

Comment/Action: We thank the reviewer for catching this inconsistency in compared quantities. Instead, we have added a direct comparison of peak tracer concentrations in CSF and parenchyma with the values by Watts et al.

- L350-352: *Different types of end nodes (root and leaf nodes) are clearly defined here. Please check everywhere in the MS these are used where relevant. For example, on Line 363, my understanding is that the end nodes referred to should be leaf nodes.*

Comment/Action: We thank the reviewer for pointing out the inconsistent use of terminal/end/leaf and root nodes. We have unified the notation, consistent with the definition of the different end node types (leaf and root) mentioned by the reviewer.

- L361: Perivascular fluid flow induced by CSF production. *I am not sure the root nodes for the PVS network are located within the SAS. Please clarify and if relevant, also clarify which value of pressure is chosen as boundary condition for these root nodes.*

Comment/Action: The reviewer is correct to point out that some of the root nodes are located (slightly) outside of the SAS. In these cases, we used the pressure value of the closest point within the SAS as the boundary condition for the pressure driven PVS flow. We remark that, as we completely removed the pressure-driven PVS flow model in the revised manuscript, this choice of boundary values is no longer used in the revised manuscript.

-SI structure *There is S1.3.1 and no S1.3.2*

Comment/Action: Right**.** As we no longer include the PVS flow induced by a constant pressure gradient (see our response to the reviewer's comment **A.a)**, we have removed said section. The same issue was present with S1.2.1 (no S1.2.2), this has been remedied by introducing a suitable subheading.

-SI L 149: *Please clarify if u_{PAR} is the intrinsic or Darcy velocity (for consistence with statement about concentration).*

Comment/Action: We have added a clarification, that u_{PAR} is indeed the Darcy velocity.